

G-CPI Forecast for High Female-Consumption
Products in Mexico

International Accelerator 2nd check-in meeting

Arthur Beltrao
(Brazil/IBGE)



Itzel Sanchez
(Mexico/INEGI)





Gender-Differentiated Price
Index and Inflation

Forecast for High Female-Consumption Products in Mexico

January 8, 2026



[Temas](#)[Programas de información](#)[Sistemas de Consulta](#)[Desarrollo social](#)[Infraestructura](#)[Acerca del INEGI](#)[Inicio](#) / [Programas de información](#) / Encuesta Nacional de Ingresos y Gastos de los Hogares Estacional, 2022

Información Demográfica y Social

[Censos y Conteos](#)[Encuestas](#)[Regulares](#)[Encuesta Nacional de
Ingresos y Gastos de los
Hogares \(ENIGH\)](#)

Encuesta Nacional de Ingresos y Gastos de los Hogares Estacional, 2022

ENIGH

La Encuesta Nacional de Ingresos y Gastos de los Hogares Estacional 2022 se llevó a cabo del 25 de diciembre de 2021 al 7 de enero de 2023. Su objetivo es proporcionar un panorama estadístico de comportamiento de los ingresos y gastos de los hogares en cuanto a su monto, procedencia y distribución a lo largo de un año; ofrecer información sobre las características ocupacionales y sociodemográficas de los integrantes del hogar; presentar datos sobre las características de la infraestructura de la vivienda y el equipamiento del hogar; identificar los periodos del año en el que los hogares muestran patrones en los que varía su gasto así como el ingreso, ya sea por trabajo, renta de la propiedad, transferencias, entre otros.

DATA SOURCE

- Dataset: **ENIGH SEASONAL 2022 (Mexico)**
- Unit of análisis: **household** `hh_key = folioviv–foliohog`

Core variables:

- **Household file:**
 - folioviv:** *Dwelling ID*
 - Foliohog:** *Household ID within dwelling*
 - Trimestre:** *Quarter (T1–T4)*
 - Factor_t:** *Quarterly expansion factor / survey weight (quarter-level weight)*
 - factor** *Annual expansion factor / survey weight (annual weight)*
- **People roster:** **población.csv**
- **Expenditure microdata:** **clave:** *item/producto code*
 - gasto_tri** = *monetary quarterly spending*
 - Gasto_nm_tri** = *non-monetary quarterly spending*
 - Total** = *gasto_tri + gas_nm_tri*

DATA/ METHOD

- ✓ Build household key and merge: quarter + weights + head sex
- ✓ Aggregate itemized spending to household totals
- ✓ Compute **weighted means** by quarter × head sex
- ✓ Estimate gaps: absolute $H - M$ and % vs women
- ✓ Approx. 95% CIs (weighted variance + effective sample size)

Progress output

Final analytic sample: 44,250 households



```
1) Data sources loaded (microdata)
- hogares: OK
- poblacion: OK
- gastoshogar: OK
- gastospersona: OK
- jefes: OK
- hh_fix_names: OK

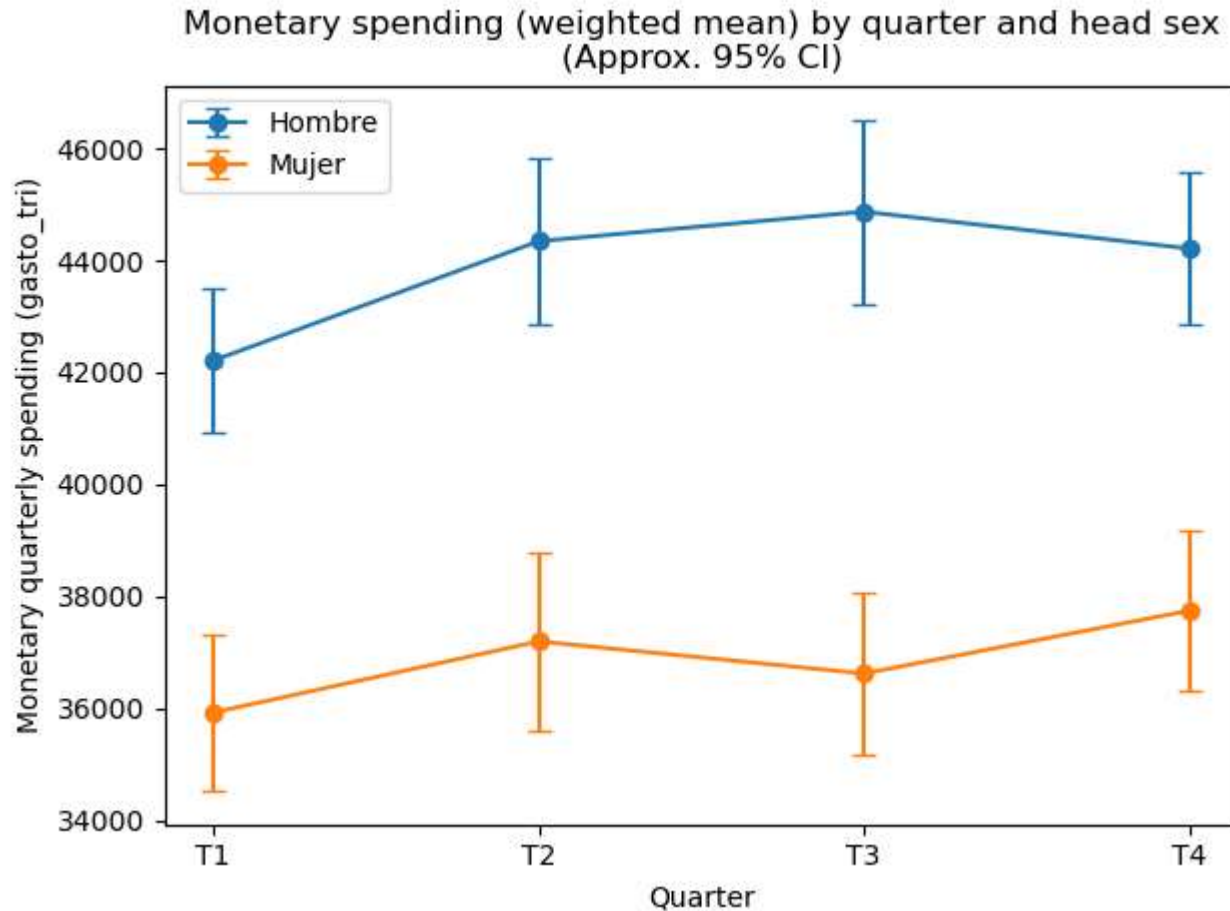
2) Dataset sizes
- hogares: 44,250 rows x 141 cols
- poblacion: 152,277 rows x 185 cols
- gastoshogar: 2,640,557 rows x 28 cols
- gastospersona: 211,022 rows x 21 cols

3) Household-level spending components (constructed)
- Unique households (hh_key): 44,250
- Definitions used:
  * Monetary (market purchases): gasto_tri -> mon_tri
  * Non-monetary (in-kind/autoconsumption): gas_nm_tri -> nm_tri
  * Total: mon_tri + nm_tri -> total_tri
- Share with monetary=0 & non-monetary>0: 0.07%
- Consistency check (total == mon+nm): 100.00% pass (should be ~100%)

4) Survey weights / quarter readiness
- hogares has factor_t (quarterly weight): True
- hogares has trimestre (quarter): True
- factor_t > 0 share: 100.00%

5) Results produced (if available)
- ci_all present. Components: ['MONETARIO', 'NO_MONETARIO', 'TOTAL']
- Monetary mean (weighted) by quarter (preview):
trimestre  Hombre  Mujer
T1 42211.49 35924.06
T2 44341.24 37194.34
T3 44869.01 36619.88
T4 44210.17 37743.11
```

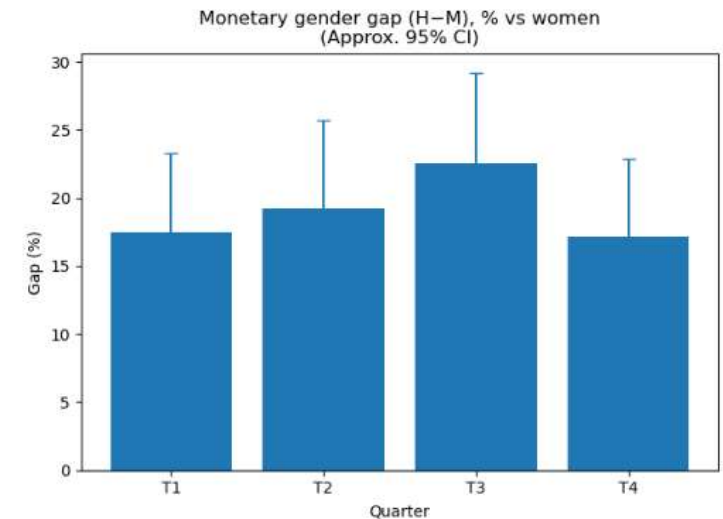
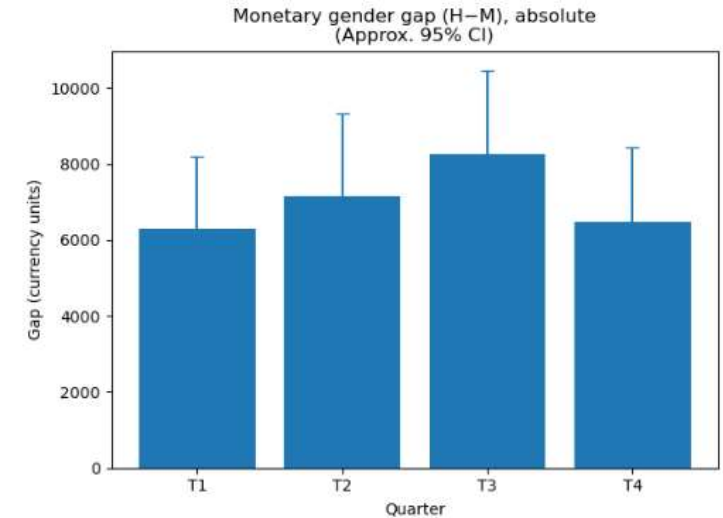
MONETARY SPENDING LEVEL BY QUARTER



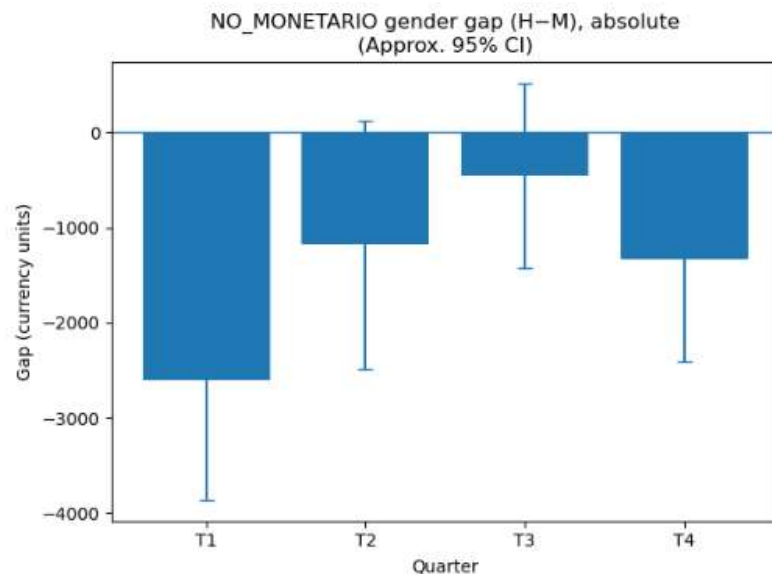
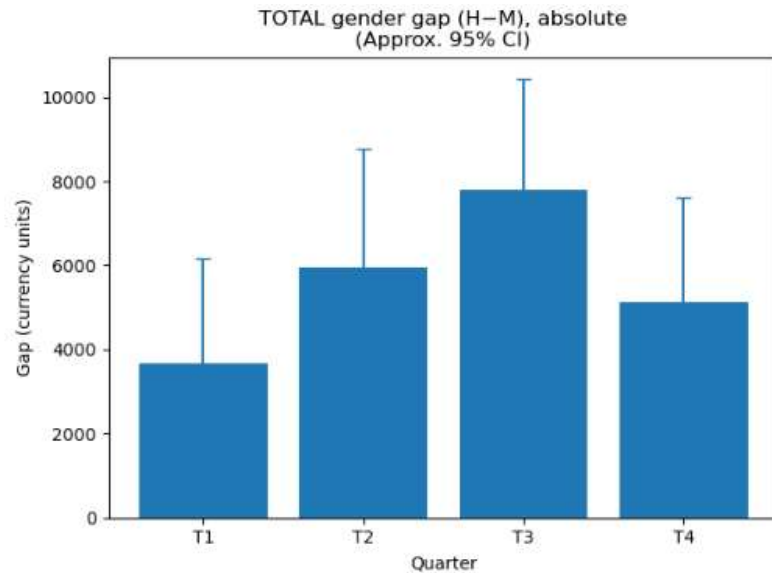
- Monetary spending shows **seasonality** across quarters
- Male-headed households have **higher monetary spending** each quarter
- CPI relevance: monetary = market purchases

MONETARY GAP BY QUARTER

- ☞ Monetary gap H-M is **positive** in all quarters
- ☞ **CI**s do not cross zero → robust difference
- ☞ Percent gap provides scale relative to women-headed households



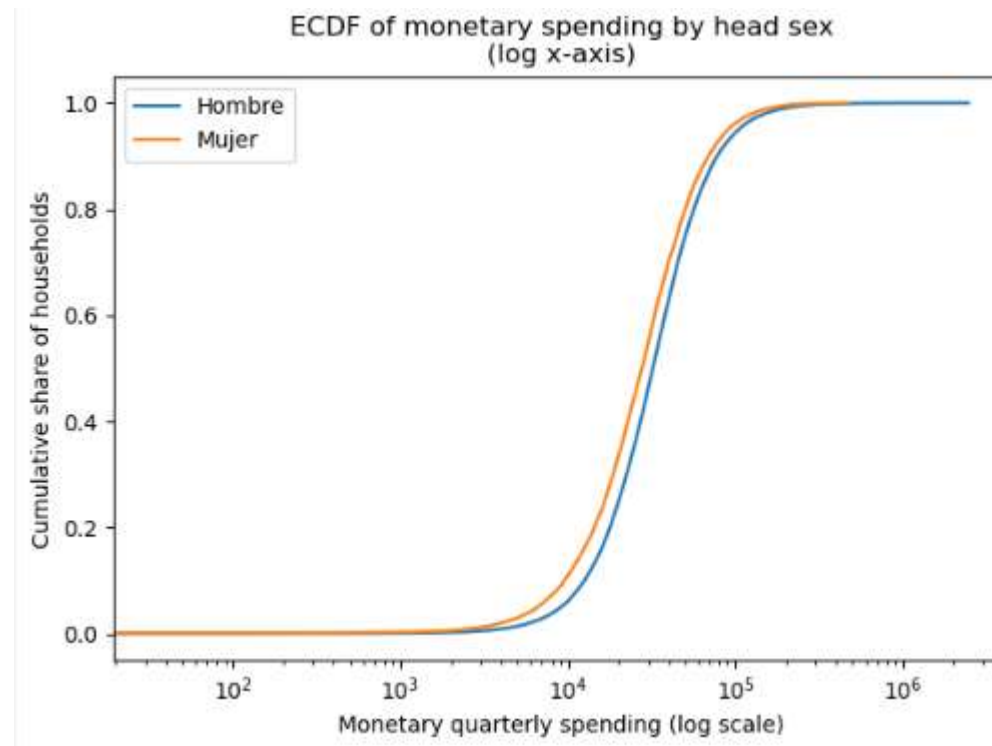
TOTAL VS NON-MONETARY GAPS



- 💰 Total gap remains **positive**
- 💰 Non-monetary gap tends to be **negative**
- 💰 Non-monetary spending **offsets** part of the total difference

Robustness: distribution evidence

- Spending is **skewed**, so averages may hide distribution shifts
- Empirical Cumulative Distribution Function (in Spanish: Función de distribución acumulada empírica) ECDF shows differences across a broad range, not only outliers



TAKEAWAYS

Key takeaways

- Monetary spending gap (H-M) is positive and robust across quarters
- Non-monetary gap tends to be negative and offsets part of the total difference.
- Seasonality is present, so quarter-aware comparisons are essential.

Next steps

- Validate monetary averages vs INEGI tabulations and confirm weight choice.
- Map item codes (clave) to CPI-style categories to build group-specific expenditure weights.
- Construct a G-CPI-style index and decompose contributions by category.

Summary table (gaps by component and quarter)

sexo	componente	trimestre	brecha_abs	brecha_pct_vs_mujer
0	MONETARIO	T1	6287.43	17.50
1	MONETARIO	T2	7146.90	19.22
2	MONETARIO	T3	8249.13	22.53
3	MONETARIO	T4	6467.05	17.13
4	NO_MONETARIO	T1	-2603.50	-16.09
5	NO_MONETARIO	T2	-1180.85	-7.45
6	NO_MONETARIO	T3	-447.99	-2.96
7	NO_MONETARIO	T4	-1333.01	-8.02
8	TOTAL	T1	3683.93	7.07
9	TOTAL	T2	5966.05	11.25
10	TOTAL	T3	7801.14	15.07
11	TOTAL	T4	5134.04	9.44

¡GRACIAS!



800 111 46 34

atencion.usuarios@inegi.org.mx



inegi.org.mx

